

Easy Schedule User Guide

V1.06

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1 Introduction

Easy Schedule is a Windows 95/NT based time of day Audio Visual control and monitoring system designed for continuous operation without the necessity for operator intervention. This manual provides a brief introduction and tutorial for using and modifying the system.

There are three essential components to the system, devices, sequences and schedules. These are introduced in the sections below, and more fully explained in later sections.

1.1 Devices

A device is a piece of equipment that can be controlled by the scheduler. A typical example would be the Electrosonic Videoserver, or an Alcorn McBride V16 show controller. The scheduler contains built in protocols for a number of devices, and provides support for most other serial or Ethernet controlled equipment via a special programmable device.

1.2 Sequences

A sequence is a time-based list of commands to send to the devices in the system. A typical sequence would be something like a 'start of day' which may require the following actions to occur...

- 1) After 1 second start playing a file on a Videoserver.
- 2) After 2 seconds turn the power on to Barco projector 1
- 3) After 3 seconds turn the power on to Barco projector 2

It can be seen from this list that an instruction in a sequence requires three pieces of information, A time, a device and an operation. The above commands could be more succinctly written as follows.

00:00:01	Videoserver	Play Chapter 1
00:00:02	Barco1	Power on
00:00:03	Barco 2	Power off

This is the format used to display sequences in the scheduler. Times in a sequence are relative to the time the sequence is started. For example a time of 00:01:00 means 1 minute from the start of sequence.

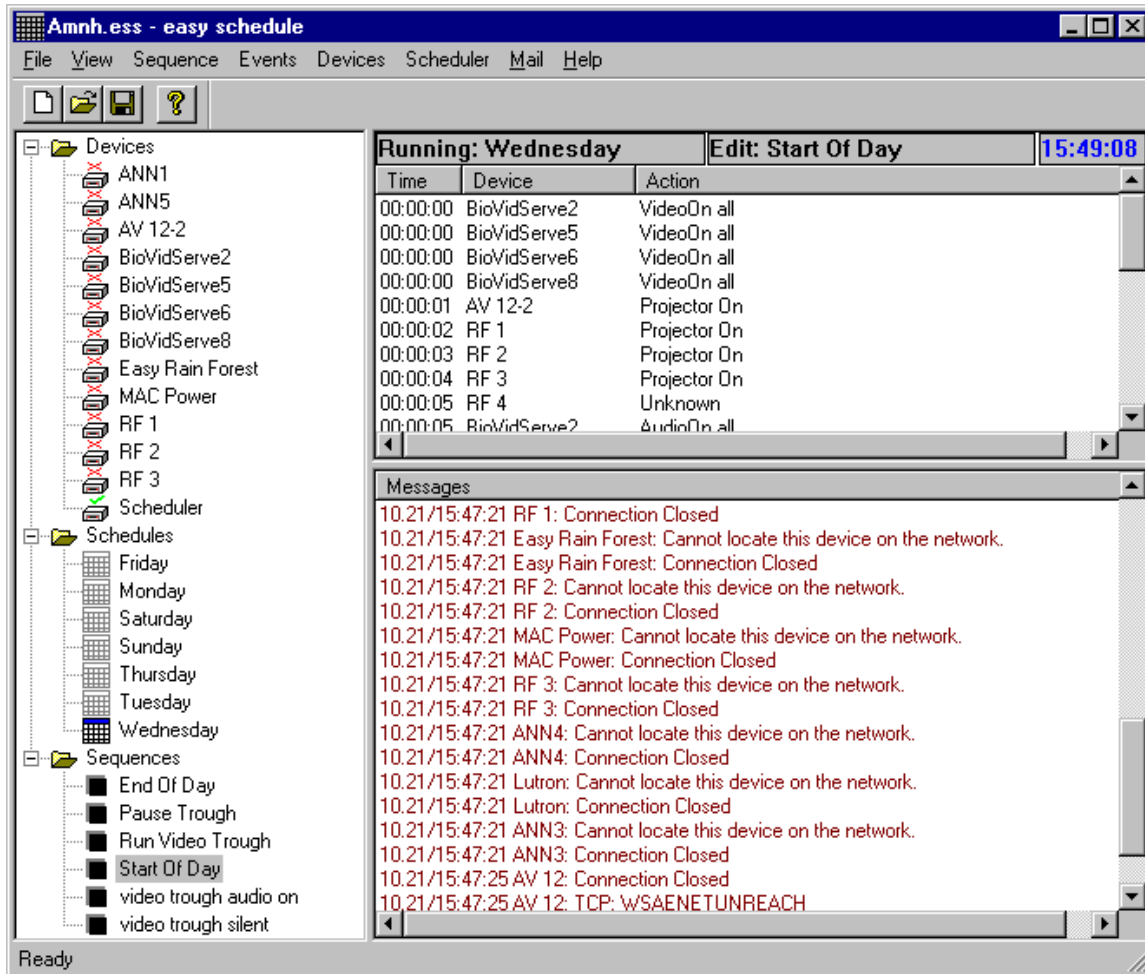
1.3 Schedule

A schedule can be thought of as a sequence where times are based on time of day. Sequences fall into two categories, general days and specific dates. General days cover the usual case where the system runs the same everyday. Specific days let the user set a schedule that occurs on a single day, December 25 for example.

The following sections describe the general concepts and operation of Easy Schedule. A device reference section explains in more detail about the commands available for the different devices.

2 The Main Display

The following screen shot shows the Easy Scheduler screen for a typical installation.



The main Easy Schedule screen is split into three main areas.

- 1) The left-hand window shows the Devices, Sequences and Schedules present in the installation. This is referred to as the Configuration Window.
- 2) The top right window is the Sequence Edit Window.
- 3) The bottom right window displays the message log. This log records everything that occurs in the Scheduler.

The amount of space allocated to each of these sections can be changed by clicking left over the dividing lines and dragging them around.

The following sections describe each of these areas in more detail. These descriptions are fairly

high level. Subsequent sections will explain how to perform specific tasks.

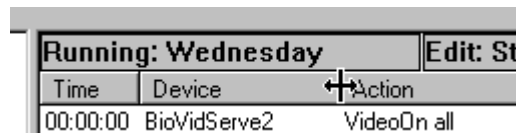
2.1 Configuration Window

The configuration window contains the setup of the installation. This includes the Devices, Sequences and Schedules. Each of these types is covered in detail in later sections.

2.2 Sequence Edit Window

The Sequence Edit Window is used to create and edit the contents of cues in both sequences and schedules. This area contains the following pieces of information

- 1) The currently running schedule is displayed in the “Running” section. This information is also replicated in the Configuration window if the Schedules tree has been expanded.
- 2) The sequence or schedule that is currently being edited is shown in the “Edit” section. As far as the editor is concerned, there is no difference between a sequence and a schedule.
- 3) The current time is shown in blue at the top right of the edit window.
- 4) The cues in the sequence. As described in the introduction there are three pieces of information required in a cue, the time, the device and the action to perform. This information is shown for each cue. The amount of information shown for each of these three fields can be altered by clicking left and dragging the divider lines on the title bar as shown in the following picture.



2.3 Message Window

The Message window is used to record everything that occurs in the Scheduling system. Color coding is used to distinguish the different types of information recorded. The following broad categories are used to distinguish the messages.

- 1) Schedule operational messages are shown in black. These messages record things such as what time the scheduler was started, what schedule file is loaded, when a sequence is started etc.
- 2) Executing Cues are shown in light gray. The Scheduler creates a log entry for every cue that is executed.
- 3) Warning messages are shown in rust. These messages show things, such as when a device disconnects.
- 4) Error messages are shown in bright Red. Error messages can be generated from the Scheduler, such as when a day is reached which does not have a schedule, or from a device driver when failures occur.

- 5) Device responses are shown in green. These messages are device driver dependant. An example message would be for a Media Matrix device reporting when a preset has completed.

The message window will display the last 1000 entries received. All log messages are also written out to disk, in a separate file for each date.

3 Devices

3.1 A brief introduction to TCP/IP

For Easy schedule to control a device there needs to be a communications link, that passes commands between the scheduler and the device. Easy schedule is designed to use TCP/IP as the communications link. TCP/IP is a set of protocols that allow two computers (or other devices) to pass data between them. It was made popular by the explosive growth of the Internet for which the protocol was originally designed. The best way to understand the concepts surrounding TCP/IP is to think about the problem that it solves using an example.

A typical network will have many different computers each running many different programs. Imagine you are sitting at a computer and wish to “talk” to a different computer. You need two pieces of information, the computer you wish to talk to and the program running on that computer. As an example from the Internet, you may wish to talk to Electrosonic’s main computer and retrieve a web page from the application that handles web page requests. In the language of TCP/IP the following notation is used...

The computer is referred to using an IP address.

The application is referred to using a PORT number.

The computer that initiates the connection (Your Computer) is called a client.

The computer that is connected to (Electrosonic’s server) is called a server.

Every computer on a network has a unique IP Address and every application running on the same computer has a unique port number (ranging from 1 to 65535). At the lowest level an IP address consists of 4 numbers between 1 and 255, written as in the following example 205.2.12.44. IP addresses can also be referenced in a more human readable form such as electrosonic.com. mnemonic names such as these are translated into their numeric equivalent by your computer.

Easy scheduler acts as a client for as many servers as are appropriate for the installation.

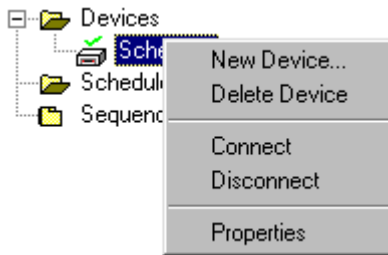
Using TCP/IP as the main communication method introduces a slight problem. How do we control RS232 based equipment such as projectors and source machines. Easy Schedule solves this problem using a very clever piece of equipment called a Port Server II. This box contains between 16 and 48 RS232 serial ports and a single TCP/IP Ethernet connection. By using this single ethernet connection Easy Schedule can talk to all 48 serial ports as if they were local to the computer.

TCP/IP, and networks in general, provides huge benefits for distributed show control systems such as Easy Schedule. Locality is effectively removed as an issue, and since networks are inherently scalable the number of devices controlled is almost limitless. If you need another 48 serial devices, just add a new Port Server to the network.

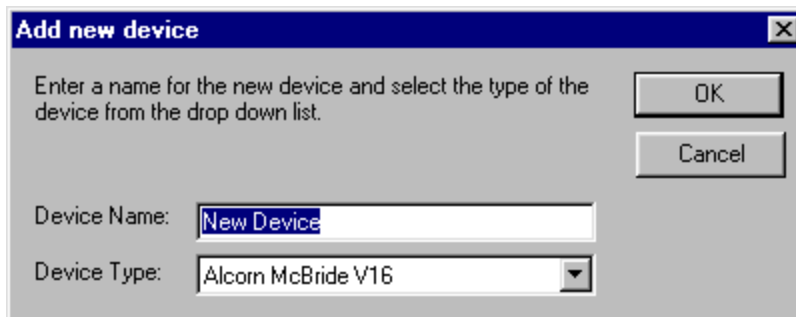
3.2 Creating a new Device

There are two ways to create a new device for use by the scheduler...

- 1) Press ALT-D on the keyboard.
- 2) Right click over any device shown in the left hand tree view and select “New Device” from the drop down menu.



Both these methods will produce the following dialog window.



Every device must have a unique name, which can be entered in the device name field. The drop down list at the bottom of the window allows the device type to be selected. When the OK button is clicked, the properties for the device are displayed. These will vary depending on the device type selected, but will usually include a tab in which the IP address and Port number for the device will be set.

The device reference at the end of this manual contains more details about the properties for individual device types.

3.3 Deleting a Device

There are two ways to delete a device from the scheduler...

- 1) Click left over the device in the Configuration window and press DELETE on the keyboard.
- 2) Right click over any device shown in the left hand tree view and select “Delete Device” from the drop down menu

Easy schedule will prompt you to make sure of your intention before removing the device.

3.4 Managing Device Connections

For Easy Schedule to be able to control a device, a TCP/IP connection needs to be established with the device. This connection is automatically made when the scheduler first starts running. If the connection is lost to the device however, it will need to be reset manually (after the reason for the failure is determined and fixed). To make the connection to all devices in the system, select “Connect to all” from the “devices” menu. This will try to re-establish communications to all devices in the system.

The connection to all devices can be broken using the “Disconnect devices” command under the “devices” menu.

The more usual situation occurs when only one or two devices are not connected. This would occur if the device was turned off and back on (or rebooted in the case of a Videosever).

Note: Occasionally the Port Server does not recognize that a connection has been lost to one of it's ports and any attempt to re-connect will fail. If this should happen you will need to kill the connection manually on the Port Server.

3.5 Understanding the Device Status

Devices in Easy Schedule can be in three different states..

3.5.1.1 *Disconnected*

A device is disconnected if Easy Schedule does not have a valid TCP/IP connection established with the device. This will be indicated by a red X by the device name in the left hand tree view.

3.5.1.2 *Connected and working*

This state is reached when Easy schedule has a TCP/IP connection to the device, and the device is functioning correctly. This state is shown by a green tick by the device name in the left hand tree view.

3.5.1.3 *Error*

This state indicates that Easy schedule has a TCP/IP connection to the device and the device has indicated an error of some sort. This will be shown by a red box with the text “ERR” , shown next to the device name in the left hand tree view.

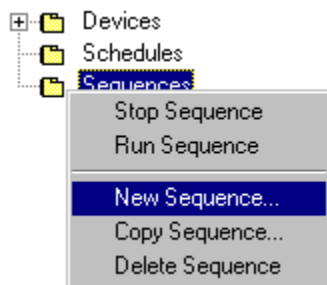
An important concept to understand is the difference between the TCP/IP connection status and the device status. A TCP/IP connection is OK if the schedule has established a connection to the computer (or Port Server) only. It does not imply that the device itself is correctly functioning. Think of a valid TCP/IP connection as the same as an RS232 cable. The fact that the cable is good does not imply that the device connected to it is working. Easy Schedule attempts to validate the operation of the device if it can, but there are many devices for which this is simply not possible.

4 Sequences

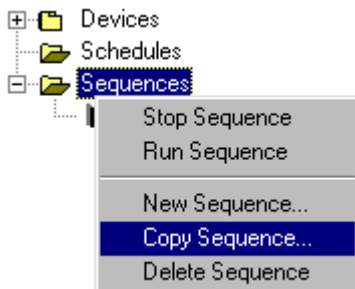
4.1 Creating a new sequence

It is possible to create a blank sequence or, more usually; copy an existing sequence. Both of these actions are achieved using a similar process.

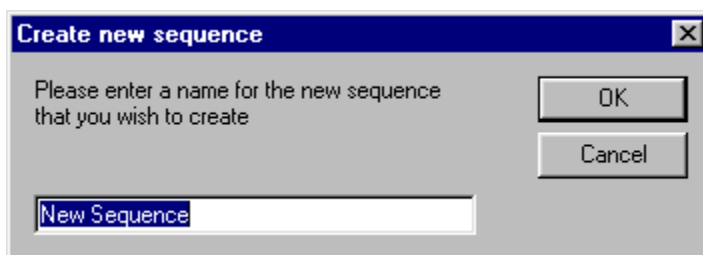
To create a new sequence, right click over any sequence in the left hand tree view, and select “New Sequence” from the drop down menu.



To copy a sequence, right click over the sequence you wish to copy and select “Copy Sequence” from the drop down menu.



Both of these actions will produce the following dialog box.

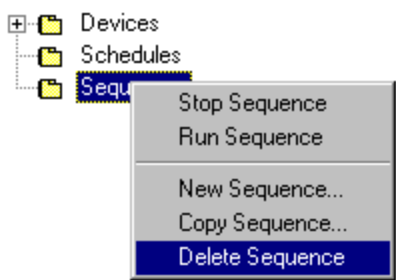


Enter a unique name for the sequence and press OK. The new sequence will become the current sequence in the event edit window.

4.2 Deleting a sequence

A sequence can be deleted in one of two ways..

- 1) Click over the sequence name in the Configuration Window and press the DELETE key on the keyboard.
- 2) Click right over the sequence name in the Configuration Window and select “Delete Sequence” from the drop down menu.



In both cases Easy Schedule will prompt to make sure you wish to proceed.

4.3 Selecting and editing a sequence

The events within a sequence can be edited by double clicking over the sequence name in the Configuration Window. This will cause the events for the sequence to be loaded into the sequence editor window. Please see the section on editing events for details about creating and editing cues.

The name of the currently edited schedule or sequence is displayed above the events list next to the word Edit:

4.4 Understanding the Sequence status icons

There are two icons used to reflect the state of a sequence. These are as follows..

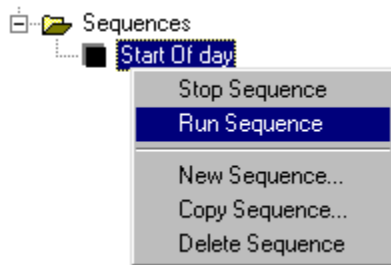
- 

 The sequence is not running
- 

 The sequence is running.

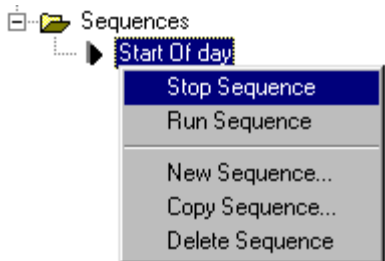
4.5 Running a Sequence

Sequences can be started manually in the following way. Right click over the sequence you wish to run and select “Run Sequence” from the drop down menu.



4.6 Stopping a Sequence

A running sequence can be stopped in the following way. Right click over the sequence you wish to stop and select “Stop Sequence” from the drop down menu.



5 Schedules

5.1 Introduction

Easy schedule manages time of day events using the concept of a day schedule. A day schedule is a sequence of actions that should occur at specific times during a day, with time 00:00:00 being midnight at the start of the day, and 23:59:59 being 1 second to midnight at the end of the day.

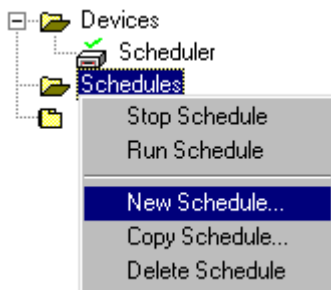
Easy schedule performs the following actions as the time crosses midnight at the start of a day.

- 1) looks for a schedule for the given date and runs it if found. This is called a specific day, “June 27 1998” for example.
- 2) If no specific date is found, the scheduler looks for a schedule with the same day of the week. For example if the day of the week is Monday, the scheduler will try to locate and run a schedule called “Monday”.
- 3) If that fails the scheduler looks for a schedule called “Daily”. If that fails no schedule is activated, and an error is reported in the log.

5.2 Creating a new Schedule

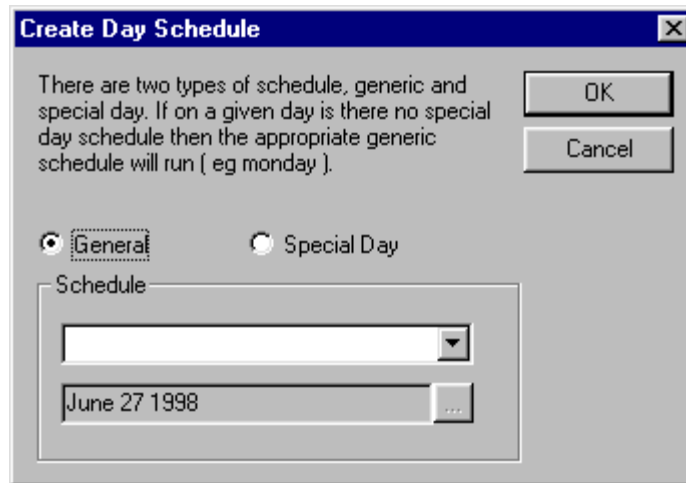
It is possible to create a blank schedule or, more usually, copy an existing schedule. Both of these actions are achieved using a similar process.

To create a new schedule, right click over any schedule in the left hand tree view, and select “New Schedule” from the drop down menu.



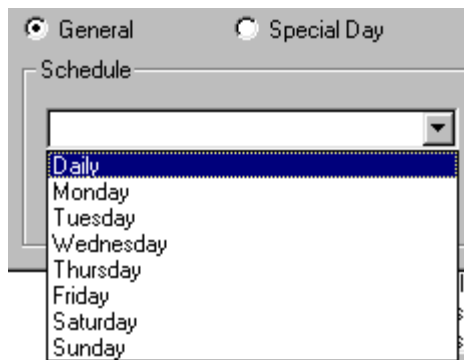
To copy a schedule, right click over the schedule you wish to copy and select “Copy Schedule” from the drop down menu.

Both of these actions will produce the following dialog box.



As mentioned above schedules can be for general days (Monday, daily etc) or for specific days (Dec 25, Jun 1 etc).

To create a schedule for a general day ensure that the General button is selected (as in the picture above) and click on the down arrow next to the blank box. This will produce a list of the general schedules that are available.



Select an entry from the list and click OK to create the schedule.

To create a schedule for a specific day, ensure that the specific day button is selected and click on the button with three small dots on. This will produce a calendar window from which a date can be selected.



After selecting a date, press Enter to accept the date or hit ESC to close the calendar without keeping the date. Once a date has been selected click OK to create the schedule. The new schedule will become the current sequence in the event edit window.

5.3 Selecting and editing a Schedule

The events within a schedule can be edited by double clicking over the schedule name in the Configuration Window. This will cause the events for the schedule to be loaded into the sequence editor window. Please see the section on editing events for details about creating and editing cues.

The name of the currently edited schedule or sequence is displayed above the events list next to the word Edit:

5.4 Deleting an existing Schedule

A schedule can be deleted in one of two ways..

- 3) Click over the schedule name in the left hand tree pane and press the DELETE key on the keyboard.
- 4) Click right over the schedule name in the left hand tree pane and select “Delete Schedule” from the drop down menu.

In both cases Easy Schedule will prompt to make sure you wish to proceed.

5.5 Suspending and resuming a Schedule

In some cases it may be necessary to suspend the currently running schedule. When a schedule is suspended no commands are sent out to devices when an event’s trigger time is reached, although the schedule is still running. This means when the schedule is resumed any events which should have fired during the suspended time will be ignored.

A schedule can be suspended by selecting “Suspend” from the “Schedule” top line menu item. This menu item is context sensitive so that the next time it is selected it will say “Resume”.

5.6 Forcing Easy Schedule to re-activate a Schedule

There may be occasions where you want to force Easy Schedule to reload the schedule that should be running on a given day. This scenario could occur when you wish to change what is happening today. Take the case where the normal end of day sequence has occurred, and the schedule has completed but there is a special event due to take place that evening. After editing the commands in the schedule so that they match the time of the special event, the schedule can be restarted using the “Activate” command under the “Scheduler” menu item.

5.7 Understanding the Schedule status Icons

There are three different icons that are used to indicate the state of a schedule. These are as follows...



The schedule is not active (either not for today or today and has completed)



The schedule is active.



The schedule has been suspended.

6 The Sequence Editing Window

6.1 Creating a new event

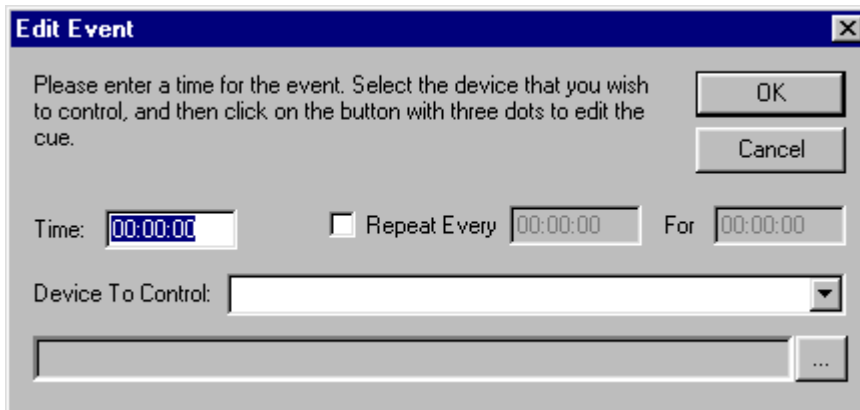
A new event can be created in one of two ways.

- 1) Press ALT-E on the keyboard.
- 2) Select the “New” command from the “Event” top line menu item.



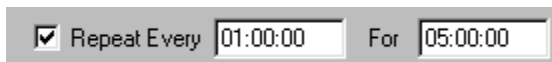
Either method will produce the event edit dialog window.

6.2 Edit Event Dialog



An event consists of three components, a time, the device that you wish to control, and the action you wish to perform on the device. Times are entered as HH:MM:SS. An event can also be programmed to repeat on a fixed period. To make an event repeat click on the “Repeat Every” button. This will activate the two other time edit windows. The left most field specifies the cycle time, and the right field specifies the length of time to repeat the event.

6.2.1.1 Example



In this example the event will repeat every one hour for five hours.

Once the time has been entered a device can be selected by clicking over the down arrow in the “Device To Control” field. After selecting a device click on the button with three dots to edit the contents of the event. This will produce a device specific dialog box for editing the action. For details of different device cues refer to the devices reference at the end of the manual.

After editing all the fields select “OK” to create the event.

6.3 Editing an existing event

The contents of an existing event can be edited by double clicking on the event. This will produce the Event edit dialog box. See the section on “creating a new event” for more details.

6.3.1 Deleting events

An event can be deleted from a sequence by clicking over it, and pressing the DELETE key on the keyboard. Easy shedule will prompt for confirmation before removing the event.

Hint: Multiple events can be selected for deletion by holding the Ctrl key down while left clicking over the events you wish to delete. Alternatively a contiguous group of events can be deleted by holding down the Shift key and clicking left over the first event and then clicking left over the last event.

7 External Triggers

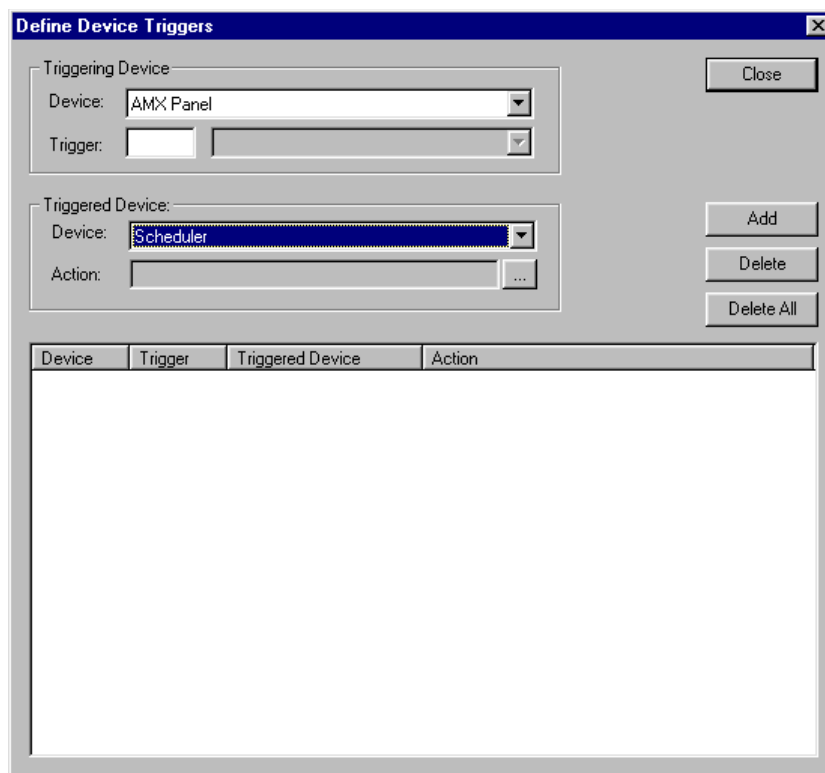
In addition to time based events, Easy Schedule can trigger an event (or run a sequence using the built in Sequence device) based on a serial message or a notification from a device. Internally serial messages are actually dealt with as notifications from the Remote trigger device type.

Triggers are entered into the system using the Device trigger dialog box. This can be found under the Sequence topline menu item.



This will produce the Define Device Triggers dialog box.

7.1 Define Triggers Dialog



A trigger basically consists of two elements, the source of the trigger and the action to carry out. These are referred as “Triggering Device” and “Triggered Device” respectively.

Triggering devices fall into two different types. Numeric triggers are from devices that send the remote trigger protocol. In this protocol a trigger is simply a number between 1 and 999. This would usually be used with an AMX touch panel. The second type of trigger is from a device that can send notifications when certain things happen. These notifications are device specific and are described in more detail in the devices reference.

7.1.1 Creating a Trigger

7.1.1.1 To create a trigger follow these steps..

- 1) Choose the device that will generate the trigger. For numeric triggers the trigger identifier should be entered as in the example below.



The image shows a dialog box titled "Triggering Device". It contains two fields: "Device:" with a dropdown menu showing "AMX Panel" and "Trigger:" with a text input field containing "512".

- 2) Create the action that is to be performed when the trigger fires. Editing an event for a trigger is exactly the same as editing an event in a sequence. For details please see the section on editing events.
- 3) Click on the “Add” button This will add the trigger to the trigger list window.

7.1.2 Deleting a trigger

A trigger can be deleted by selecting it from the list and left clicking the delete button. To select a trigger, click left over the device name in the left most column of the list. Easy schedule will prompt for confirmation before removing the trigger.

7.1.3 Deleting all triggers

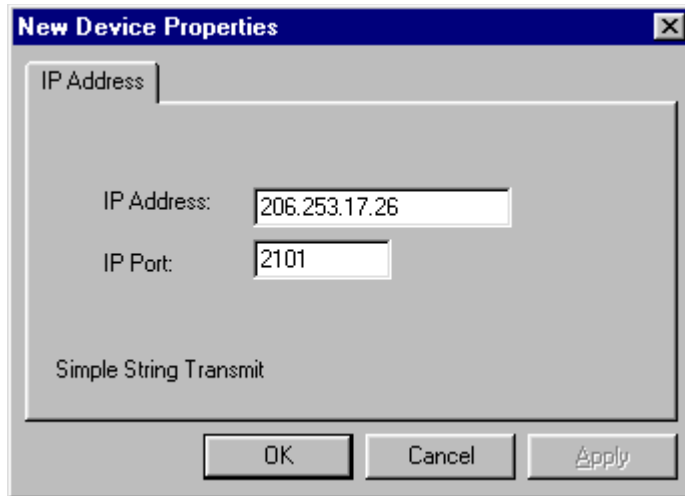
The entire contents of the trigger list window can be erased by clicking over the “Delete All” button. Easy Schedule will prompt for confirmation before removing all the triggers.

Click Close when you have finished editing the trigger lists.

8 Device Reference

8.1 Common Properties

Almost all devices require a TCP/IP address. This is entered in the TCP/IP properties page.



For details about IP addresses and Port numbers refer to the devices section.

8.2 Aurora

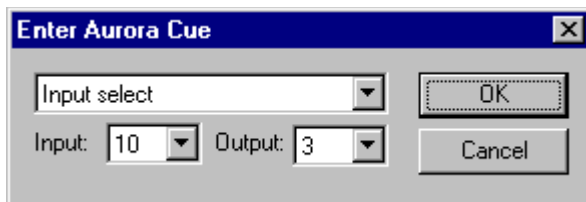
The Aurora is a “video mixer” usually used in security camera applications. It allows up to sixteen different video feeds to be monitored on a single display.

8.2.1 Properties

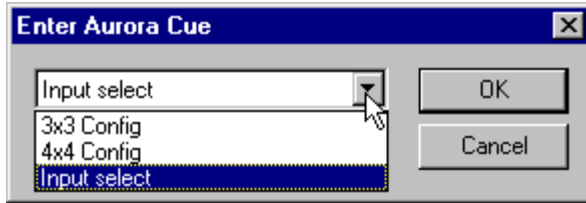
The Aurora has no custom properties.

8.2.2 Commands

After selecting the edit cue button for an Aurora device, the following dialog box is displayed.



The Aurora supports three different commands, which can be selected by left clicking over the top “List Box” as illustrated in the following screen shot.



8.2.2.1 3x3 Config

This command causes the Aurora to switch to a 3x3 configuration, where the screen is split into 9 different sources. Selecting this command will cause the Input and Output windows to gray out, indicating they cannot be edited.

8.2.2.2 4x4 Config

This command causes the Aurora to switch to a 4x4 configuration, where the screen is split into 16 different sources. Selecting this command will cause the Input and Output windows to gray out, indicating they cannot be edited.

8.2.2.3 Input select

This command is used to change which inputs appear in the 3x3-configuration screen. The Output field selects the segment of the output to change, according to the following diagram..

1	2	3
4	5	6
7		

with an extra option that allows all segments to be selected. The Input field selects which input to display in the segment.

8.3 Barco Projectors

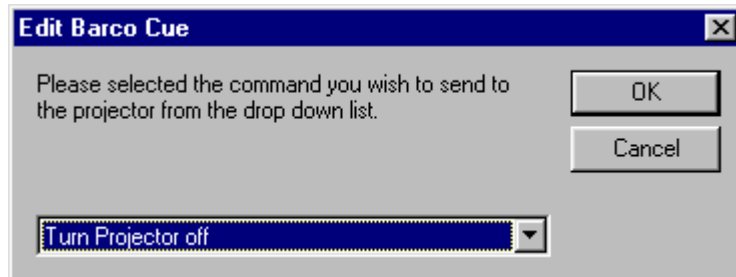
This device can be used to send commands to the Barco family of LCD projectors.

8.3.1 Properties

There are no custom properties for Barco Projectors.

8.3.2 Commands

There are currently two supported commands available for the Barco device, turn power on and turn power off.



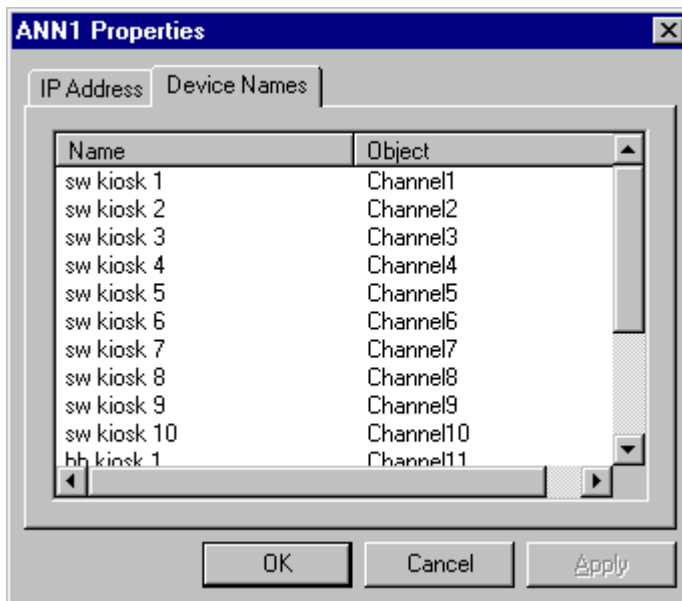
Click on the down arrow to select a command and press OK to accept.

8.4 DataProbe Annunciator unit

The Dataprobe Annunciator unit provides monitoring over electrical signals, and can provide warnings when failures occur. This device is often linked to a video sync detector providing a warning upon loss of video signal to an exhibit. Each Annunciator can monitor 16 signals.

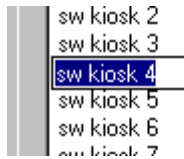
8.4.1 Properties

In addition to the standard TCP/IP address, the names for the 16 channels can be entered from the following properties page.



A name can be changed either by clicking left over a name, and then clicking left again, or

clicking left over the name and pressing the “INSERT” key on the keyboard. This will produce a small edit window in which the new name can be entered. Press “ENTER” to accept the new name.



8.4.2 Commands

The Annunciator is a monitoring device only, and does not support any schedule commands.

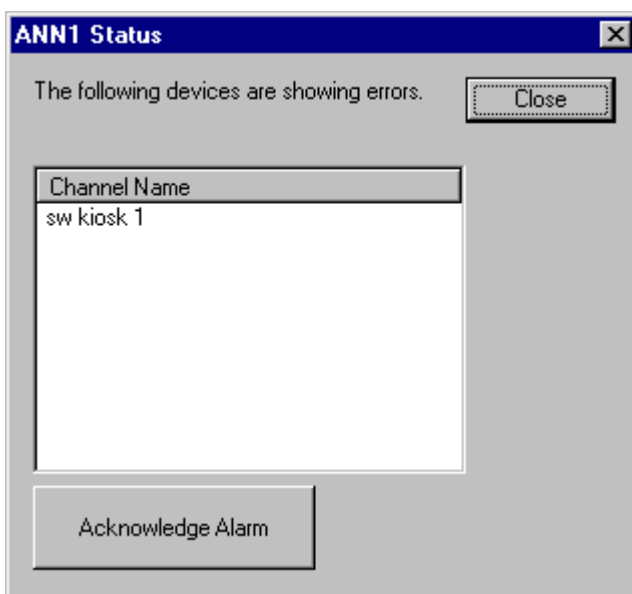
8.4.3 Handling Alarms

This section describes how alarms are reported in Easy Schedule, and how to deal with them.

When an Annunciator unit detects an error, it enters an alarm mode. This unit itself shows this alarm with a flashing red light on the channel in question and a possible audio alert if enabled. The failure is also detected by Easy Schedule, which creates entries in the log indicating the time of the failure and the device that failed. At this point the failing device will show the “error” indication in the device tree, as shown by the following picture.



While the error condition is present, double clicking over the device will bring up a status window instead of the device properties dialog. For an Annunciator unit this status window is as follows.



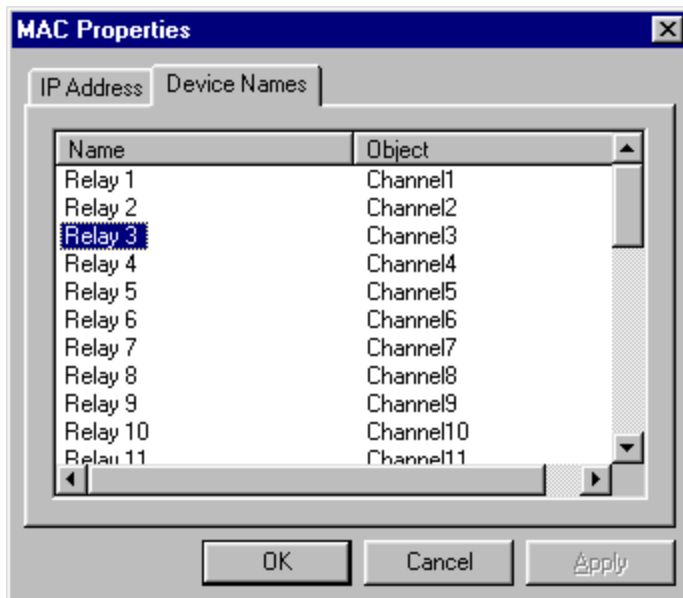
This dialog shows a list of devices that have failed. The Acknowledge Alarm button at the base of the window is used to tell the Annunciator that the alarm has been noticed. At this point the alarm light will stop flashing and the audible warning will stop. The alarm condition will still be present in both the Annunciator and Easy Schedule, until the source of the problem is fixed. When the problem is rectified Easy schedule will report the time in the log and clear the error condition on the devices tree.

8.5 DataProbe MAC32 power control unit

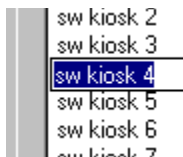
The DataProbe MAC32 provides 32 Relays which can be used to control power management units.

8.5.1 Properties

In addition to the TCP/IP address, the names for the 32 channels can be entered from the following properties page.

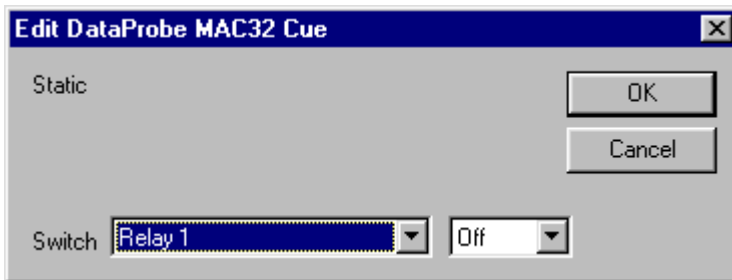


A name can be changed either by clicking left over a name, and then clicking left again, or clicking left over the name and pressing the “INSERT” key on the keyboard. This will produce a small edit window in which the new name can be entered. Press “ENTER” to accept the new name.



8.5.2 Commands

The Dataprobe MAC32 device supports two commands, turn the Relay off, and turn the Relay On (These are actually inverted inside the driver – since the Power modules used with the MAC32 use the relay energized state to turn power off).



To create the event select the relay you wish to change and select either on or off from the drop list.

8.6 Pioneer CAC V180M

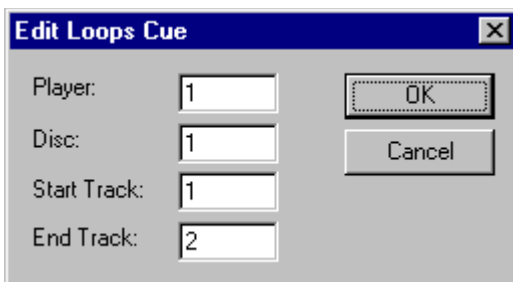
The Pioneer CAC V180M is a RS232 serially controllable CD multichanger. The player can play 1 of 15 disks using 3 separate 5 disc cartridges. The device driver for this device allows a group of tracks to be looped repeatedly.

8.6.1 Properties

There are no custom properties for the

8.6.2 Commands

Selecting the cue edit button for this device will produce the following dialog box.



To create a looping cue, change the Disc, Start Track and End Track fields to the required settings and click OK.

Note that the Player No. is included for compatibility with a previous version. This should always be left at a value of 1.

8.7 Simple String Transmit

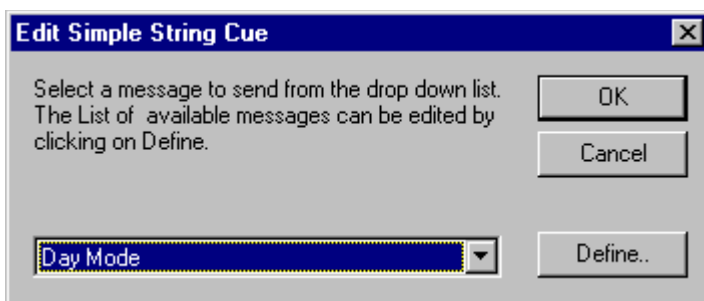
The simple string transmit device is used to control devices for which there is no standard device driver in Easy Schedule. It allows messages to be created containing a mix of Hex and ASCII data, which can be sent out from a cue in a sequence.

8.7.1 Properties

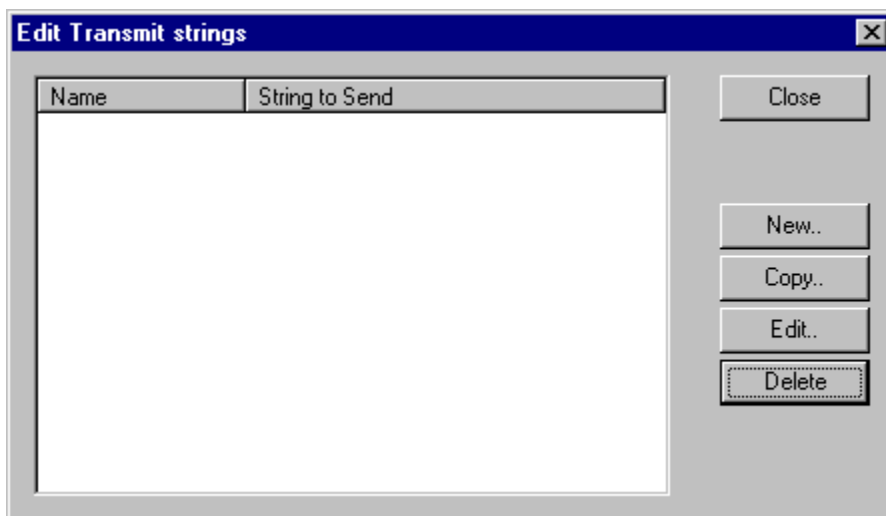
There are no custom properties for this device.

8.7.2 Commands

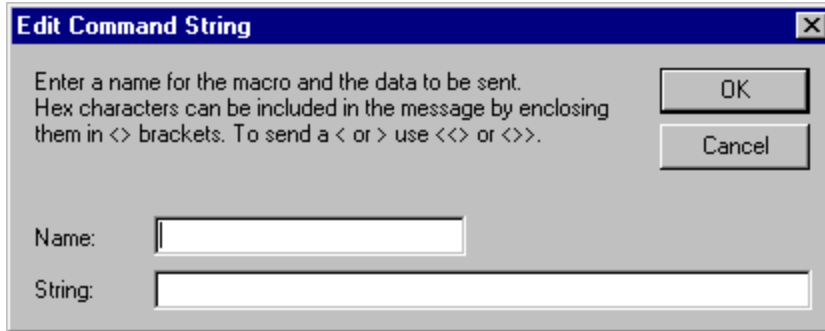
The Cue edit dialog box for a Simple String Device is as follows..



The drop down list (highlighted in blue) contains the names of strings that have already been defined. To create a cue, simply select the required string and click on OK. To define new strings, click left over the “Define” button. This will produce the “Edit Transmit Strings” dialog box.



A string can be added to the table by clicking on the “New” button. This will produce the “Edit Command String” dialog box.



This dialog contains two edit fields, for entering a message and mnemonic name used to reference the message. Messages can be any length containing any ASCII character. Non-printable characters can be embedded in the string by enclosing a hex value in a pair of angle brackets. The following table shows some example strings and the hex data that will be sent out.

String	Hex output
<01>R<02><17><03>	01 52 02 17 03
HELLO<13><10>	48 45 4C 4C 4F 13 10
<<>H<>>	3C 48 3E

Click “OK” to create the message and add it to the message table.

In many cases, a string will need to be created that is very similar to a string that is already in the message table. The “Copy” button is provided on the Edit Transmit strings dialog box for this purpose. To copy an existing message, highlight the message by clicking left over the name and then click on the “Copy” button. This will produce the “Edit Command String” dialog from where the name and message can be edited.

An existing string can be edited by highlighting the name, and clicking left over the “Edit” button. This will produce the “Edit Command String” dialog from where the message can be edited. (Note that in the edit mode it is also possible to change the Name for the string. This will result in a new entry being created in the table).

Strings can be deleted by highlighting the name, and clicking left over the “Delete” button. A confirmation window will appear to prevent accidental deletion. More than one name can be highlighted by holding down the CTRL key while using the left mouse button to select a name.

8.8 Remote Trigger

The Remote trigger device provides a remote control interface for use by external equipment. This allows third party equipment, such as a touch panel, to control the operation of Easy Schedule.

The trigger protocol is very simple, consisting of a square open bracket, followed by a three digit ascii number, followed by a square close bracket. For example, [123] sends the trigger 123 to the scheduler.

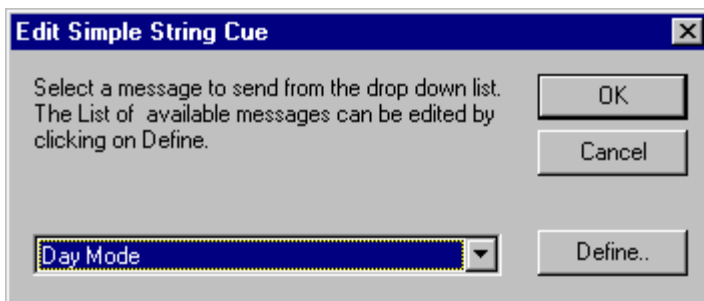
8.8.1 Properties

There are no custom properties for this device.

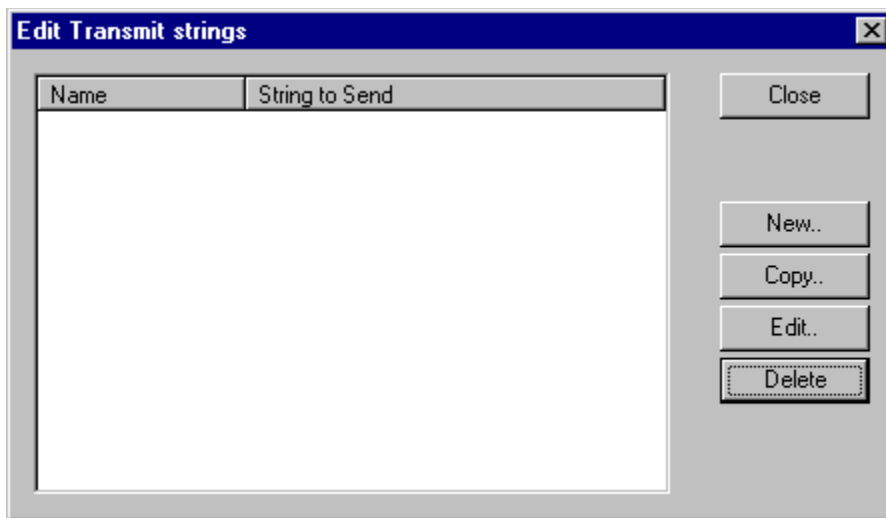
8.8.2 Commands

As well as providing trigger firing from the device, the Remote Trigger driver allows strings to be defined that can be sent to the device. A typical use for this feature is to send status information to a touch screen so that as sequences are run in the Scheduler, the panel shows any modes or settings that are in effect. Strings are defined through the normal cue edit mechanism.

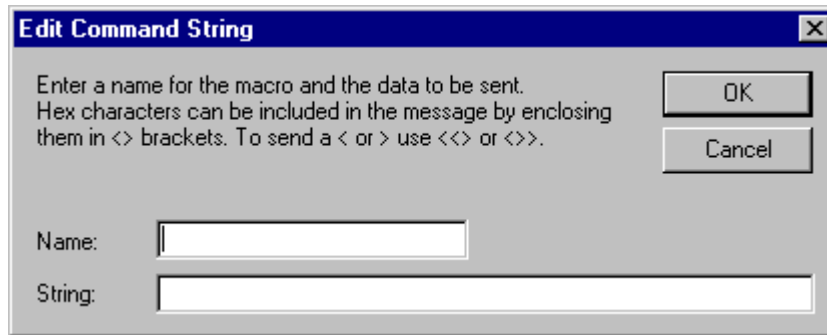
The Cue edit dialog box for a Remote Trigger Device is as follows..



The drop down list (highlighted in blue) contains the names of strings that have already been defined. To create a cue, simply select the required string and click on OK. To define new strings, click left over the "Define" button. This will produce the "Edit Transmit Strings" dialog box.



A string can be added to the table by clicking on the “New” button. This will produce the “Edit Command String” dialog box.



This dialog contains two edit fields, for entering a message and mnemonic name used to reference the message. Messages can be any length containing any ASCII character. Non-printable characters can be embedded in the string by enclosing a hex value in a pair of angle brackets. The following table shows some example strings and the hex data that will be sent out.

String	Hex output
HELLO	48 45 4C 4C 4F
HELLO<13><10>	48 45 4C 4C 4F 13 10
<<>H<>>	3C 48 3E

Click “OK” to create the message and add it to the message table.

In many cases, a string will need to be created that is very similar to a string that is already in the message table. The “Copy” button is provided on the Edit Transmit strings dialog box for this purpose. To copy an existing message, highlight the message by clicking left over the name and then click on the “Copy” button. This will produce the “Edit Command String” dialog from where the name and message can be edited.

An existing string can be edited by highlighting the name, and clicking left over the “Edit” button. This will produce the “Edit Command String” dialog from where the message can be edited. (Note that in the edit mode it is also possible to change the Name for the string. This will result in a new entry being created in the table).

Strings can be deleted by highlighting the name, and clicking left over the “Delete” button. A confirmation window will appear to prevent accidental deletion. More than one name can be highlighted by holding down the CTRL key while using the left mouse button to select a name.

8.9 Media Matrix

This device provides control over presets stored in a Peavey Media Matrix audio processing system.

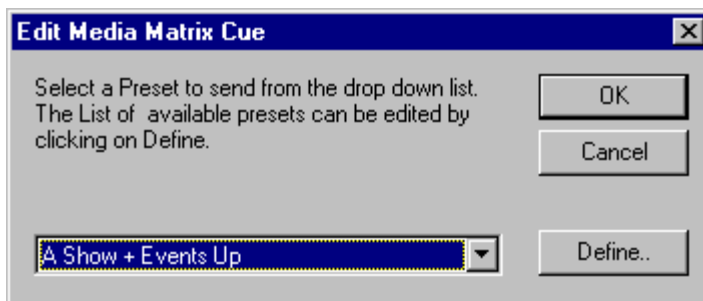
8.9.1 Properties

There are no custom properties for the Media Matrix device.

8.9.2 Commands

The Media Matrix device provides two basic commands, turn preset on, and turn preset off. To make programming easier, these presets are given names, which can then be quickly incorporated into a cue.

The Cue edit dialog for a Media Matrix device appears as follows..



The required preset can be selected from the drop down “list box”. In the example above the preset “A Show + Events Up” has been selected.

The first time this dialog is used the drop down window will be blank because there will be no presets defined. To define presets, left click on the “Define” button. This will produce the following dialog box.

The following pieces of information are required for a preset.

- 1) A name. This should be something descriptive such as “Volume Low”.
- 2) The Preset No. This value is obtained from the Pasha.ini file on the Media Matrix.
- 3) On or Off. Media Matrix presets can be turned on or off. The usage of these two options is dependent on the Media Matrix programming.

To create a new preset, enter the information described above and click left over the “Add” button. This will create a new entry in the preset table. The following picture shows an example screen, with two presets defined.

Name	Variable	State
Mute All Outputs	5	On
UnMute All Outputs	5	Off

Entries can be deleted from the preset table by highlighting an entry and clicking left over the “Delete” button. A confirmation message will appear to prevent accidental deletion.

A “Delete All” button is also provided, allowing the entire table to be deleted. As for “Delete”, confirmation is required to prevent accidental deletion.

8.10 Electrosonic Videoserver

The Electrosonic Videoserver is a 8 or 12 channel MPEG2 digital video and audio delivery system, capable of independent channel control or synchronized playback of groups of channels.

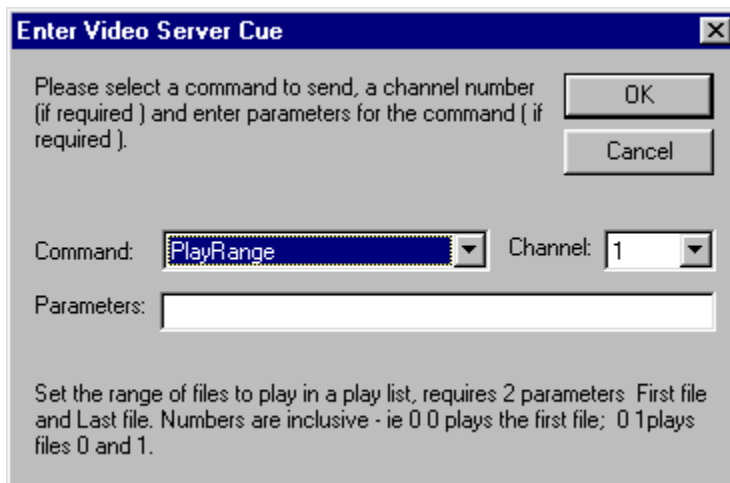
For more information please refer to the Videoserver documentation.

8.10.1 Properties

There are no custom properties for the Videoserver device.

8.10.2 Commands

Videoserver commands are entered using the following dialog box.



Video server commands consist of the command, the affected channel (or all channels), and any optional parameters. As the commands are changed the bottom section of the window will provide help on the required parameter list.

The Videoserver supports the following commands.

8.10.2.1 *PlayList*

This command is used to play the files selected by the previous PlayRange command.

8.10.2.2 *PlayRange*

This command is used to select a range of files to play in a playlist. It requires two parameters, the first file to play, and the last file to play. These are inclusive, i.e. a parameter list of 0 1 will select files 0 and 1.

8.10.2.3 Play

This command is used to play the currently selected file on the requested channel.

8.10.2.4 Stop

This command is used to stop the currently playing file on the requested channel.

8.10.2.5 Seek

This command is used to search to a given frame of the mpeg file. The command accepts a frame number as the only parameter.

8.10.2.6 Video Off

This command is used to set the video output to black.

8.10.2.7 Audio Off

This command is used to mute the audio.

8.10.2.8 Video On

This command is used to put the mpeg video to the output.

8.10.2.9 Audio On

This command is used to put the mpeg audio to the output.

8.11 Alcorn McBride V16

This device driver is used to control Alcorn McBride's range of equipment, using the Alcorn 8 bit protocol. Easy Schedule supports a subset of this protocol, which allows a variable to be assigned to a value in the V16. The V16 protocol also supports a reporting mechanism that allows the V16 to send back status messages to Easy Schedule. These messages can also be used as triggers.

8.11.1 Properties

The properties page for an Alcorn McBride V16 device is used to define new triggers, which can be used to control sequences within Easy Schedule. For a full description of Triggers, please refer to the External Triggers section of this manual.

The following screen shot shows the properties page for a typical V16 device.

The screenshot shows the 'Interactive V16-1 Properties' dialog box with the 'Triggers' tab selected. It contains a table of triggers and fields for adding or deleting them.

Id	Level	Description
1	Notify	Main show started
2	Warning	Main show audio failed
3	Error	Alarm triggered

Fields above the table: Id: 3, Level: Error, Add button, Name: Alarm triggered, Delete button.

Buttons at the bottom: OK, Cancel, Apply.

A trigger has three pieces of information

- 1) The Id is the trigger identifier. This value is passed in by the V16 as part of the message protocol. See the “Trigger Protocol” section for details.
- 2) A Level indicator. This specifies how the message should be handled by the Scheduler logging and fault reporting mechanism. All messages are kept in the log files and screen log display, but the Scheduler also provides remote E-mail capability, which is controlled by the level settings. There are currently three levels, Notify, Warning and Error. Notify should be used for logging normal operational messages. Warning should be used for operations that may require investigation but are not major failures that disrupt the operation of an exhibit. Errors are usually reserved for major failures that require immediate attention.
- 3) Description. This is used to identify the trigger in the “Edit Triggers” dialog. Note that this name does not appear in the log files. The log file text is provided as part of the message from the V16.

To create a new Trigger in the table, simply fill in the three pieces of information in the appropriate fields and click left over “Add”. This will add the trigger to the table. If the trigger Id is already used in the table, a warning message will appear ensuring that Id’s cannot be accidentally replaced.

To delete a Trigger, highlight the trigger by clicking left over it’s Id, and then click “Delete” to remove it. A confirmation message will appear to prevent accidental deletion.

8.11.2 Trigger Protocol

This section describes the protocol that an Alcorn McBride V16 controller uses to communicate with Easy Schedule. These messages will not be acknowledged by the scheduling system.

The message structure follows the Electrosonic ES4000 protocol.

SOH	Type	Header	STX	Data	ETB	ETX
^A	E	n1	^B	message	^W	^C
01	45		02		17	03

Where ...

n1

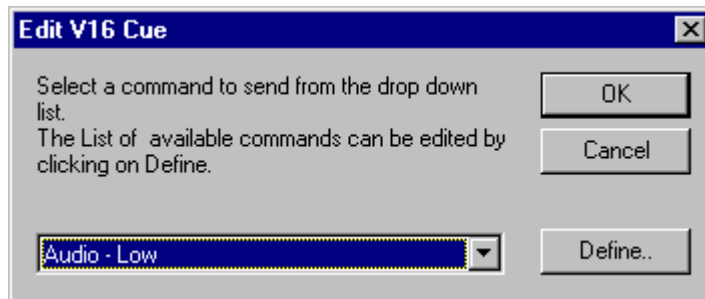
This number is used to identify the message. The value is in ascii, for example message 23 will be sent as 0x32 0x33.

Message

A textual description of the event that occurred. The string should not be null terminated and can be any length.

8.11.3 Commands

The following screen shot shows the V16 device cue edit window.



This dialog box allows a predefined command to be selected from the drop down list box. In the example above, the command “Audio - Low” has been selected. Commands can be defined by left clicking over the “Define” button, producing the following dialog.

Define V16 Commands

Name:

Variable: Value:

OK Cancel

Name	Variable	Value
------	----------	-------

Add Delete Delete All

Each Command contains three pieces of information.

- 1) The Name is a mnemonic reference, which will appear in the cue.
- 2) Variable refers to a variable in the Alcorn McBride V16. Note that in the V16 programming system, called WinScript, variables start from 1 whereas in the Alcorn McBride 8 bit protocol, variables start at 0. For this reason the variables used in Easy Schedule should be 1 less than variables used in WinScript. For example, consider a sequence defined in WinScript which is triggered when variable 30 is equal to 10. To remotely start this sequence from Easy Schedule, send a value of 10 to variable 29.
- 3) Value is the value that the variable will be set to.

To create a new command fill in the three fields described above and left click on the “Add” button. This will create a new entry in the table. Note that it is possible to create two different names for the same variable/value combination.

A command can be deleted by highlighting the command name in the list and clicking left on the “Delete” button. A confirmation window will appear to prevent accidental deletion.

A “Delete All” button is also available for clearing out the entire contents of the table. As for “Delete”, confirmation will be required.

8.12 Show Clock

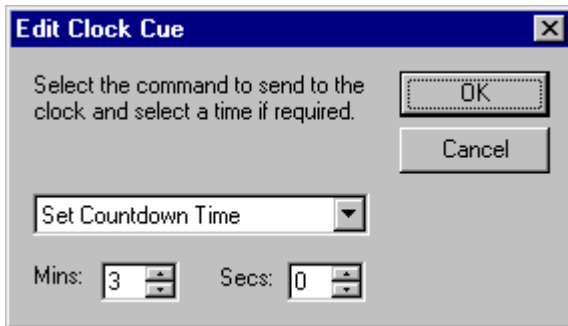
This device driver is used to control the custom Large Clock developed by Harris Productions. This device is only in use in one installation of Easy Schedule, and was developed as custom work for that site. The programming information is included here for completeness.

8.12.1 Properties

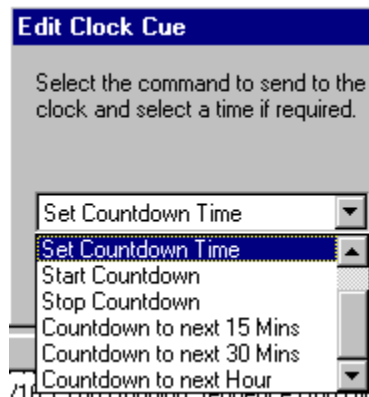
There are no custom properties for this device.

8.12.2 Commands

The cue edit window for the Show Clock device is as follows..



To change the command type, click left over the command window and select an entry from the drop down “List Box”.



The Mins. and Secs. edit windows will activate and de-activate depending on the command selected. The following commands are available.

8.12.2.1 Turn Clock On

This command causes the clock to turn on, i.e. illuminate the digits.

8.12.2.2 Turn Clock Off

This command causes the clock to turn off, i.e. the digits will not be illuminated.

8.12.2.3 Start Countdown

This command causes the countdown to begin from the preloaded time to 0.

8.12.2.4 Stop Countdown

This command causes the countdown to stop at the current time.

8.12.2.5 Set Countdown time.

This command is used to set the countdown starting point to a specific time. This is the only command that uses the Mins. And Secs. fields.

8.12.2.6 Countdown to next 15 Mins.

This command is used to load the countdown starting point to the time required to countdown to the next ¼ hour. For example if the cue is executed at 10:04:00 the countdown time will be set to 11 minutes and 0 seconds.

8.12.2.7 Countdown to next 30 Mins.

This command is used to load the countdown starting point to the time required to countdown to the next 1/2 hour. For example if the cue is executed at 10:04:00 the countdown time will be set to 26 minutes and 0 seconds.

8.12.2.8 Countdown to next hour.

This command is used to load the countdown starting point to the time required to countdown to the next hour. For example if the cue is executed at 10:04:00 the countdown time will be set to 56 minutes and 0 seconds.

8.12.3 Typical Countdown sequence

The following screen shot illustrates the typical sequence of cues used to start a countdown on the clock. This screen is taken from a sequence designed to show a three minute countdown.

Time	Device	Action
00:00:00	Show clock	Set Countdown for 03:00
00:00:00	Show clock	Start Countdown
00:00:01	Show clock	Turn On

8.13 Scheduler

8.13.1 Properties

The scheduler device does not have any properties.

8.13.2 Commands

The scheduler is a special built in device that provides control over the general operation of Easy Schedule. The following operations are available..

8.13.2.1 Call Sequence

This command can be used to start a sequence running. Sequences are not interruptible, so if the sequence is already running this command will have no effect.

8.13.2.2 Stop Sequence

This command can be used to stop a running sequence. If the sequence is not running, the command will be ignored.

8.13.2.3 Disable Device TX

This command is used to disable the transmission of data to a remote device. This might be used in the situation where a complex multi sequence day schedule is in operation and you wish to lock out a device without stopping other sequences which refer to the device.

8.13.2.4 Disable Device RX

This command is used to prevent a device from firing triggers in the scheduler. A common use for this command would be to disable an interactive touch panel while a large show is running.

8.13.2.5 Enable Device TX

This is the opposite operation to disable Device TX. It is used to re-enable communications to a device.

8.13.2.6 Enable Device RX

This is the opposite operation to disable Device RX. It is used to re-enable triggers from the device.

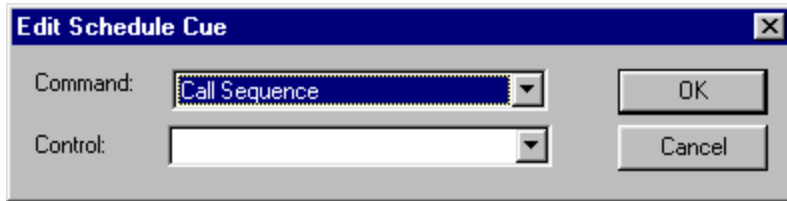
8.13.2.7 Disable Day Schedule

This command is used to suspend the current day schedule. This might be used to temporarily interrupt the normal operation of the day, to run some special event sequences.

8.13.2.8 Enable Day Schedule

This is the opposite command to Disable Day Schedule. It will re-activate the day schedule at the current point in time. That is if the schedule was disabled at 12pm and activated at 1pm, the next event to execute will be at 13:00:01. All events between 12 and 1 will be ignored.

When a scheduler cue is created the following dialog box appears for editing the action.



The “command” drop down box is used to enter the action to perform, and the “Control” drop down box is used to enter any necessary action specific data. In the case of the device RX/TX control this would be a device name, and in the case of the Sequence start/stop commands this would be a sequence name. The available options in the “Control” list box changes depending on the Command selected.

Click “OK” to accept the changes or “Cancel” to abort the edit.

8.14 Knox Switcher

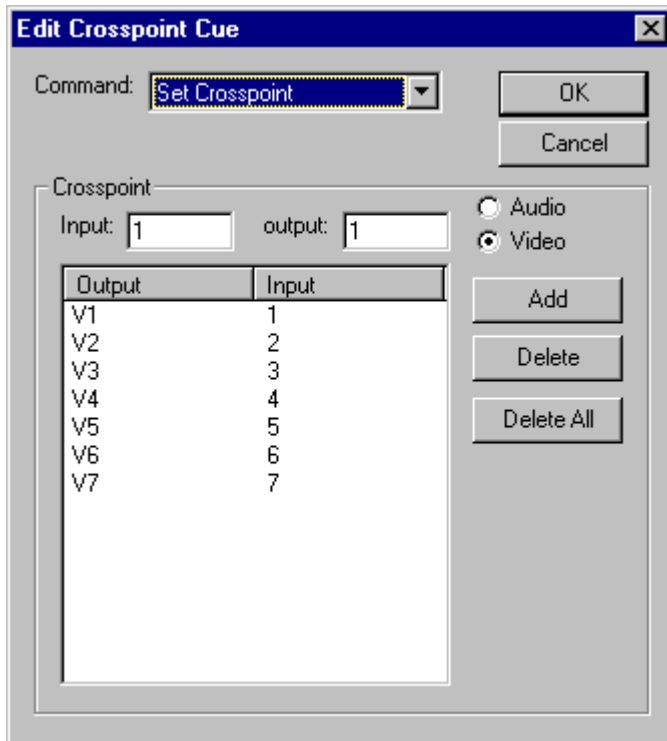
This device driver provides control over the Knox range of Video/Audio switchers. Provision is provided for separate switching of Video and Audio. In addition, multiple crosspoints can be set from a single cue.

8.14.1 Properties

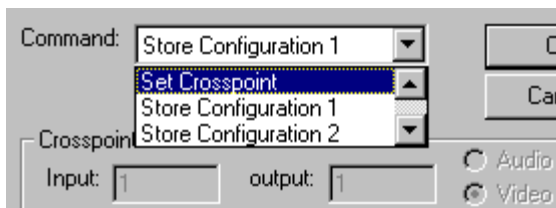
There are no custom properties for the Knox switcher device.

8.14.2 Commands

The cue edit dialog for the Knox switcher is as follows..



The Knox device supports a number of different commands, which can be selected from the drop down “list box”.



The following commands are available.

8.14.2.1 Store Configuration 1

This command causes the Knox switcher to save the current Crosspoint settings internally to memory 1. These can be recalled later to provide a very fast full channel switch.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.2 Store Configuration 2

This command causes the Knox switcher to save the current Crosspoint settings internally to

memory 2. These can be recalled later to provide a very fast full channel switch.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.3 Store Configuration 3

This command causes the Knox switcher to save the current Crosspoint settings internally to memory 3. These can be recalled later to provide a very fast full channel switch.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.4 Recall Configuration 1

This command causes the Knox switcher to recall the Crosspoint settings from memory 1.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.5 Recall Configuration 2

This command causes the Knox switcher to recall the Crosspoint settings from memory 2.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.6 Recall Configuration 3

This command causes the Knox switcher to recall the Crosspoint settings from memory 3.

Selecting this command will cause the Crosspoint portion of the window to gray out, because they are not needed for this cue type.

8.14.2.7 Set Crosspoint

This command is used to set a number of Audio and Video crosspoints on the switcher.

To add a Crosspoint to the cue, the following steps are required.

- 1) Select the required Crosspoint in the input and output edit windows.
- 2) Highlight the radio button for either Video or Audio as required.
- 3) Click left over the “Add” Button.

This will cause the Crosspoint to be added to the Crosspoint table. The following screen shot shows an example cue with two entries, an audio switch connecting input 3 to output 1, and a video switch connecting input 7 to output 4.

Crosspoint

Input: output:

Output	Input
A1	3
V4	7

Entries can be deleted from the Crosspoint table by highlighting an entry and clicking left over the “Delete” button. A confirmation message will appear to prevent accidental deletion.

A “Delete All” button is also provided, allowing the entire table to be deleted. As for “Delete”, confirmation is required to prevent accidental deletion.

8.15 Flipper Controller

The flipper device is a device driver written to control a specific piece of custom software designed to display sports results on a large mechanical “train station” flipper style display. The device details are included in this manual for completeness.

8.15.1 Properties

There are no custom properties for this device type.

8.15.2 Commands

The Cue edit dialog box for the Flipper Controller is as follows.

Select Flipper command

Select a command from the drop down list. To create a new Look, click on Define.

There are three commands that can be sent to the flipper control software.

- 1) Disable Schedule. This command stops the internal scheduler from sending messages to the displays. This would be used in situations where the flipper should be stopped, during an event for example.
- 2) Enable Schedule. This command re-enables the internal scheduler.
- 3) Send a “look” to the display. This may be used for showing special messages during the day.

Custom messages can be created by left clicking on the “Define” Button. This will produce the “Flipper Controller Cue” dialog box.

The flipper system contains 4 separate boards, each showing 50 characters. The following steps are needed to create a new look for the four boards.

- 1) Enter a name for the look. This name will appear in the “Flipper command” dialog box.
- 2) Ensure that “Board 1” is showing in the drop list at the top of the window.
- 3) Enter the required text for the 6 lines on board 1. Note that this text will be centered when it is displayed on the flipper board.
- 4) Select “Board 2” from the drop list.
- 5) Enter the required text for the 6 lines on board 2.
- 6) Repeat steps 4 and 5 for the remaining boards.
- 7) When all the text has been entered, click left on “Add” to create the new look.

A existing look can be edited by double clicking over it’s name in the “Looks” list at the bottom

of the window. After editing, use “Add” to update the look. Note that a prompt message will appear confirming you wish to update the look.

A look can be deleted by highlighting it’s name in the “Looks” window and left clicking over the “Delete” button. A confirmation window will appear to prevent accidental deletion.

8.16 Winwall

This device can be used to remotely control Electrosonic’s Worksurface control system, called Winwall. Winwall provides a simple remote control protocol allowing different effects to be started and stopped.

The Winwall remote control interface supports triggering when specific effects finish.

8.16.1 Properties

The Winwall properties page is used to create triggers which occur when Winwall effects reach the end of their timeline. The Trigger properties page is shown below.

The image shows a 'Winwall Properties' dialog box with a 'Triggers' tab selected. Inside the tab, there are three input fields: 'Id' with the value '1', 'Notify' with a dropdown menu showing 'Notify', and 'Effect Name' with an empty dropdown. To the right of these fields are three buttons: 'Add..', 'Delete', and 'Refresh'. Below these fields is a table with three columns: 'Id', 'Level', and 'Description'. The table is currently empty. At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Apply'.

Id	Level	Description
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A Winwall trigger contains three pieces of information..

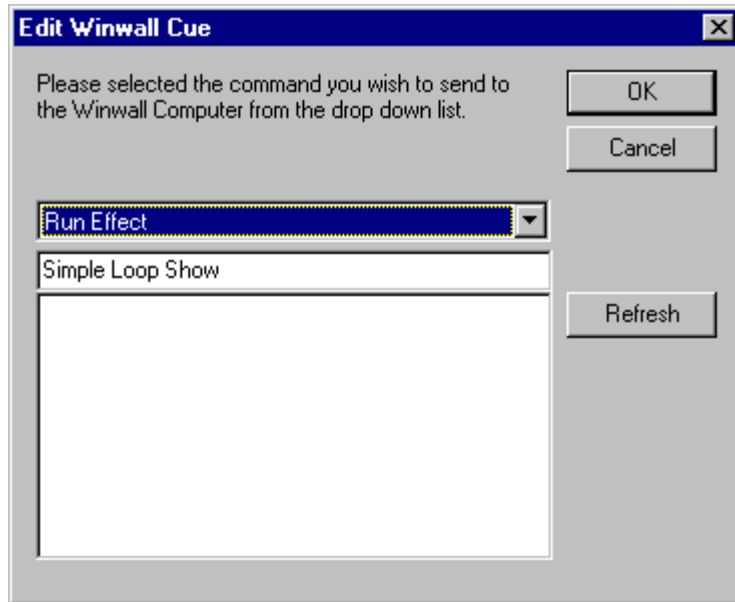
- 1) The id is used to identify the trigger. It should be different for each trigger.
- 2) The severity of the trigger. This field is included for compatibility with other devices, which can report major failures. For a Winwall device, this should be left on Notify.
- 3) The effect name. This effect must complete for the trigger to fire.

To create a new trigger these three pieces of information should be entered in the appropriate fields and the “Add” button selected. If the Scheduler has a connection to the Winwall software (i.e. the Winwall device has a green tick next to it in the Device tree) clicking on Refresh will

retrieve the list of effects in the current Winwall show.

8.16.2 Commands

The Winwall device supports two commands, run an effect and stop the currently running effect. The following screen shot shows the Winwall cue edit dialog.



The command is selected using the drop list (showing “Run Effect” in the example screen above.). The Run Effect command requires an effect name to be entered in the edit box directly below the command. The effect name can be selected from the effects list, or entered manually. If a connection is present to the Winwall software, a list of effects in the current Winwall show can be retrieved by clicking on the “Refresh” button. After a short delay the effects list will fill with names.